Medical Assistant Project

Post Graduate Program in Artifical Intelligence and Machine Learning

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Contents / Agenda

Executive Summary

Healthcare professionals face increasing challenges in accessing accurate, timely medical information due to the growing volume of clinical data, leading to delays in decision-making and increased diagnostic risk. To address this, we developed an AI-powered medical assistant leveraging **Large Language Models (LLMs)**, progressing from a baseline model to prompt-engineered and ultimately a **Retrieval-Augmented Generation (RAG)** architecture.

Initial LLM configurations demonstrated strong natural language capabilities and baseline medical understanding but lacked depth, clinical rigor, and actionable guidance. Enhancements through prompt engineering improved structure, readability, and practical usability, enabling more comprehensive and clinically aligned responses.

The RAG-based solution integrates trusted medical knowledge from the **Merck Manual of Diagnosis and Therapy**, using vector embeddings and semantic retrieval (top-k documents) to ground responses in authoritative sources while minimizing hallucinations. A strict system prompt enforces reliance on retrieved content and ensures transparency when information is unavailable.
This evolution significantly improves response accuracy, relevance, and clinical reliability, enabling faster, evidence-based decision-making, promoting standardized care, and enhancing patient outcomes.

Finally, the tuned RAG solution significantly enhances accuracy, relevance, and clinical reliability, enabling faster, evidence-based decision-making, supporting standardized care delivery, and improving patient outcomes while maintaining consistency and explainability.

Business Problem Overview and Solution Approach

Problem Overview:

Healthcare professionals face significant challenges in accessing accurate, up-to-date medical information quickly due to the overwhelming volume of clinical data and research. This information overload can delay decision-making, increase the risk of diagnostic errors, and reduce efficiency in time-sensitive situations. There is a need for an intelligent system that can rapidly retrieve and synthesize trusted medical knowledge to support timely, evidence-based clinical decisions and improve patient outcomes

Solution Approach:

To address information overload and enable faster, evidence-based clinical decisions, the solution leverages a **Retrieval-Augmented Generation (RAG)** architecture that combines intelligent search with generative AI.

Trusted medical knowledge from the **Merck Manual of Diagnosis and Therapy** is ingested, cleaned, and **chunked** into smaller sections. These are converted into **vector embeddings** using transformer-based models and stored in a **vector database** for efficient similarity-based retrieval.

When a query is submitted, it is embedded and matched using **cosine similarity** to retrieve the most relevant, clinically validated content. A **large language model (LLM)** then synthesizes this information into a concise, clinically meaningful response, improving speed and reducing cognitive load.

To ensure reliability, an **LLM-as-a-Judge** framework evaluates outputs on:

- **Grounding** (alignment with retrieved content)
- **Relevance** (fit to the clinical query)

This reduces hallucinations and improves response accuracy.

Question Answering using LLM

Model Considerations

Model Selection

- **Repository**: Hugging Face
- **Model name**: "TheBloke/Mistral-7B-Instruct-v0.2-GGUF“
  - The Bloke: publisher (mid-size LLM)
  - 7B: 7 billion parameters
  - GGUF: Format used by llama.cpp for local use
- **Model basename**: "mistral-7b-instruct-v0.2.Q6_K.gguf"

Model Execution

- **Model:** Llama
- **n_ctx:** 2300 (context window)
- **n_gpu_layers**: 36 (GPU speed)
- **n_batch:** 512 (throughput)

Parameters

- **max_tokens**: 125
- **temperature**: 0 (0: fully deterministic - 1: creative)
- **top_p**: 0.95 (nucleus sampling)
- **top_k**: 50 (50 most likely words)

- This model configuration leverages a locally deployed **Mistral 7B Instruct** model optimized for **deterministic and consistent output generation**.
- The setup prioritizes **reliability over creativity** by using a temperature of 0, ensuring repeatable responses for accurate prompt evaluation and comparison.
- Performance is enhanced through GPU layer offloading and batch optimization, while the chosen quantization balances model quality and computational efficiency.
- Overall, this configuration is designed to isolate prompt effectiveness and model behavior, enabling structured analysis of LLM responses

Question Answering using LLM

Query 1: What is the protocol for managing sepsis in a critical care unit?
Llama.generate: prefix-match hit \n\nSepsis is a life-threatening condition that can arise from an infection, and it requires prompt recognition and aggressive management in a critical care unit. The f ollowing are general steps for managing sepsis in a critical care unit:\n\n1. Early recognition: Recognize the signs and symptoms of sepsis early and initiate treatme nt as soon as possible. Sepsis can present with various clinical features, including fever or hypothermia, tachycardia or bradycardia, altered mental status, respirato ry distress, and lactic acidosis.\n2. ABCs

Strengths	Opportunities
• Natural language works well. • The model correctly identifies Sepsis as life threatening. • Provides core / baseline medical concepts. • Strong semantic understanding. • The model shows solid baseline medical knowledge encoding	• Response stays high level. • Response lacks clinical depth • Response does not provide a structured protocol • The model explains what 'sepsis' is instead of what was asked to do ('…protocol for managing sepsis in a critical care unit')

Question Answering using LLM

Query 2: What are the common symptoms for appendicitis, and can it be cured via medicine? If not, what surgical procedure should be followed to treat it?

Llama.generate: prefix-match hit
\n\nAppendicitis is a medical condition characterized by inflammation of the appendix, a small pouch-like structure that extends from the large intestine. The sympto ms of appendicitis can vary from person to person, but some common signs include:\n\n1. Abdominal pain: The pain is typically located in the lower right side of the abdomen and may be constant or come and go. It may start as a mild discomfort that worsens over time.\n2. Loss of appetite: People with appendicitis often lose th eir appetite due to abdominal pain and nausea

Strengths	Opportunities
• Natural language works well. • The model correctly identifies Appendicitis as inflammation of the appendix • Provides core / baseline medical concepts. • Correctly identifies key symptoms.	• Response doesn’t include answer to the following queries: 1. Can it be treated with medication? 2. What surgical procedure should be followed? • Response lacks clinical depth. • Response is not actionable.

Question Answering using LLM

Query 3: What are the effective treatments or solutions for addressing sudden patchy hair loss, commonly seen as localized bald spots on the scalp, and what could be the possible causes behind it?
Llama.generate: prefix-match hit \n\nSudden patchy hair loss, also known as alopecia areata, is a common autoimmune disorder that affects the hair follicles. It can result in round or oval bald patc hes on the scalp, but it can also occur on other parts of the body such as the beard area, eyebrows, and eyelashes.\n\nThe exact cause of alopecia areata is not kn own, but it's believed to be related to a problem with the immune system. Some possible triggers for this condition include stress, genetics, viral infections, and cert ain medications.

Strengths	Opportunities
• Natural language works well. • The model correctly identifies links patchy hair loss to alopecia areata • Provides core / baseline medical concepts. • Good answer for general comprehension.	• Response fails to answer the full question: 1.Treatment or solutions • Answer lacks clinical rigor / robustness. • Response is incomplete and lacks depth. • Response is not actionable.

Question Answering using LLM

Query 4: What treatments are recommended for a person who has sustained a physical injury to brain tissue, resulting in temporary or permanent impairment of brain function?

Llama.generate: prefix-match hit
\n\nA person who has sustained a physical injury to the brain tissue may require various treatments depending on the severity and location of the injury. Here are some common treatments that may be recommended:\n\n1. Emergency care: In case of a traumatic brain injury (TBI), it is essential to seek emergency medical attention as soon as possible. The primary goal of emergency care is to prevent further damage to the brain, stabilize vital signs, and manage any life-threatening conditions.\n2. Medications: Depending on the symptoms, healthcare professionals may prescribe medications to manage various conditions associated with a

Strengths	Opportunities
• Correctly emphasizes seeking emergency care immediately • Provides core / baseline medical concepts. • Good answer for general comprehension.	• Response fails to answer the full question: 1.Treatment or solutions • Answer lacks clinical rigor / robustness. • Response is incomplete and lacks depth. • Response is not actionable.

Question Answering using LLM

Query 5: What are the necessary precautions and treatment steps for a person who has fractured their leg during a hiking trip, and what should be considered for their care and recovery?
Llama.generate: prefix-match hit \n\nFirst and foremost, if you suspect that someone has fractured their leg while hiking, it's essential to ensure their safety and prevent further injury. Here are some necessary precautions:\n\n1. Keep the person calm and still: Encourage them to remain as still as possible to minimize pain and prevent worsening the injury.\n2. Assess the situation: Check for any signs of shock, such as pale skin, rapid heartbeat, or shallow breathing. If you notice these symptoms, seek medical help immediately.\n3. Immobilize the leg: Use a splint, sl

Strengths	Opportunities
• Natural language works well. • The model correctly interprets world emergency scenario (fracture during hiking) • Provides core / baseline medical concepts. • Good answer for general comprehension.	• Response fails to answer the full question: 1.Treatment or solutions • Answer lacks clinical rigor / robustness. • Response is incomplete and lacks depth. • Response is not actionable.

Question Answering using LLM w/. Prompt Engineering

Great Learning
POWER AHEAD

Model Considerations

Model Selection
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- **top_p**: 0.95 (nucleus sampling)
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Prompt Engineering

```
system_prompt = """
You are a highly advanced AI Medical Assistant that provides accurate, clear, and helpful medical information
based on trusted medical knowledge such as medical textbooks, clinical guidelines,
and reliable healthcare sources.
Provide medical information in the following sections:

Overview:
Causes:
Symptoms:
Diagnosis:
Treatment:
Prevention:

Keep explanations clear, concise, and medically accurate.
If you do not know the answer, say you do not know.
"""
```

- This model configuration leverages a locally deployed Mistral 7B Instruct model optimized for deterministic and consistent output generation.
- In addition, a **domain-specific system prompt** defines the model as an AI Medical Assistant, enforcing structured responses and grounding outputs in trusted medical knowledge.
- Overall, this configuration is designed to maximize consistency, reduce hallucination risk, and standardize output structure, enabling accurate evaluation of prompt effectiveness and dependable delivery of medical information.

Question Answering using LLM w/. Prompt Engineering

Query 1: What is the protocol for managing sepsis in a critical care unit?
Llama.generate: prefix-match hit \n\nOverview:\nSepsis is a life-threatening condition caused by the body's response to an infection. It can lead to tissue damage, organ failure, and even death if not treated promptly and effectively. In a critical care unit, sepsis management involves a multidisciplinary approach that includes rapid identification, hemodynamic support, antibiotic therapy, and intensive care.\n\nCauses:\nSepsis is caused by the body's response to an infection, which can be bacterial, viral, or fungal in origin. The infection can occur anywhere in the body, such as the lungs, urinary tract, or abdomen. In a critical care unit, patients are often at higher risk for sepsis due to underlying health conditions and invasive procedures.\n\nSymptoms:\nSymptoms of sepsis can vary widely depending on the severity of the infection and the patient's overall health status. Common symptoms include fever or hypothermia, rapid heart rate, rapid breathing, confusion or altered mental status, and low blood pressure. In

Strengths	Opportunities
• The response follows a clear medical framework (Overview, Causes, Symptoms, Diagnosis, Treatment, Prevention), improving readability and alignment with clinical thinking. • The answer addresses multiple dimensions of sepsis management demonstrating breadth expected in clinical summaries.	• The response lacks detailed ICU-specific interventions • The 125-token cap restricts the ability to fully elaborate on complex protocols, leading to oversimplification. • The response does not clearly sequence actions which is critical in emergency/ICU settings. • Some explanations remain high-level and could benefit from more precise terminology

Question Answering using LLM w/. Prompt Engineering

Query 2: What are the common symptoms for appendicitis, and can it be cured via medicine? If not, what surgical procedure should be followed to treat it?

Llama.generate: prefix-match hit
\n\nAppendicitis is a medical condition characterized by inflammation of the appendix, a small tube-shaped organ located in the lower right side of the abdomen. Th e symptoms of appendicitis can vary from person to person, but some common signs include:\n\n1. Abdominal pain: The pain typically starts as a mild discomfort a round the navel or in the lower right abdomen. It may then move to the right lower quadrant and become more severe over time. The pain is often described as sha rp, crampy, or constant.\n2. Loss of appetite: People with appendicitis may lose their appetite due to abdominal discomfort or nausea.\n3. Nausea and vomiting: Vo miting is a common symptom of appendicitis, especially in the later stages of the condition.\n4. Fever: A fever of 100.4°F (38°C) or higher may be present in some cases of appendicitis.\n5. Diarrhea or constipation: Some people with appendicitis experience diarrhea, while others have constipation.\n6. Rebound tenderness: W hen pressure is applied to the

Strengths	Opportunities
• The response adheres to the predefined sections • Correctly identifies appendicitis symptoms and distinguishes between medical vs.surgical treatment pathways. • Clearly communicates that surgery (appendectomy) is the standard of care, while acknowledging limited scenarios where antibiotics may be considered.	• The 125-token cap restricts the ability to fully elaborate on complex protocols, leading to oversimplification. • The response would benefit from greater clinical specificity, urgency framing, and procedural detail.

Question Answering using LLM w/. Prompt Engineering

Query 3: What are the effective treatments or solutions for addressing sudden patchy hair loss, commonly seen as localized bald spots on the scalp, and what could be the possible causes behind it?

Llama.generate: prefix-match hit
\n\nSudden patchy hair loss, also known as alopecia areata, is a common autoimmune disorder that affects the hair follicles. It can result in round or oval bald patc hes on the scalp, but it can also occur on other parts of the body such as the beard area, eyebrows, or eyelashes.\n\nThe exact cause of alopecia areata is not kno wn, but it's believed to be related to a problem with the immune system. Some possible triggers for this condition include stress, genetics, viral infections, and certai n medications.\n\nThere are several treatments that have been shown to be effective in addressing sudden patchy hair loss:\n\n1. Corticosteroids: These are anti-i nflammatory drugs that can help reduce inflammation and suppress the immune system's attack on the hair follicles. They can be applied topically or taken orally, d epending on the severity of the condition.\n2. Minoxidil: This is a medication that has been shown to promote hair growth in some people with alopecia areata. It wo rks by increas

Strengths	Opportunities
• The response effectively identifies multiple potential causes for patchy hair loss. • The response provides a range of treatment options • The response is practical and actionable.	• The response lacks detailed treatment protocols and diagnosis methods. • No reference to clinical guidelines or evidence-best practices limiting credibility • Some explanations remain high-level and could benefit from more precise terminology

Question Answering using LLM w/. Prompt Engineering

Query 4: What treatments are recommended for a person who has sustained a physical injury to brain tissue, resulting in temporary or permanent impairment of brain function?

Llama.generate: prefix-match hit
\n\nA person who has sustained a physical injury to the brain tissue may require various treatments depending on the severity and location of the injury. Here are s ome common treatments that may be recommended:\n\n1. Emergency care: In case of a traumatic brain injury (TBI), it is essential to seek emergency medical atte ntion as soon as possible. The primary goal of emergency care is to prevent further damage to the brain, stabilize vital signs, and manage any life-threatening cond itions.\n2. Medications: Depending on the symptoms, healthcare professionals may prescribe medications to manage various conditions associated with a brain inj ury, such as pain, swelling, seizures, or infections.\n3. Surgery: In some cases, surgery may be necessary to remove blood clots, repair skull fractures, or relieve pr essure on the brain.\n4. Rehabilitation: Rehabilitation is an essential component of treatment for brain injuries. It may include physical therapy, occupational therap y, speech and languag

Strengths	Opportunities
• Emphasizes urgency of emergency care and stabilization • Correctly identifies common treatment categories	• Lack of clinical depth – No severity differentiation • No mention of guidelines • The 125-token cap restricts the ability to fully elaborate on complex protocols, leading to oversimplification. • Lacks step by step treatment sequencing

Question Answering using LLM w/. Prompt Engineering

Query 5: What are the necessary precautions and treatment steps for a person who has fractured their leg during a hiking trip, and what should be considered for their care and recovery?

```
Llama.generate: prefix-match hit
\n\nFirst and foremost, if you suspect that someone has fractured their leg while hiking, it's essential to ensure their safety and prevent further injury. Here are some necessary precautions:\n\n1. Keep the person calm and still: Encourage them to remain as still as possible to minimize pain and prevent worsening the injury.\n2. Assess the situation: Check for any signs of shock, such as pale skin, rapid heartbeat, or shallow breathing. If you notice these symptoms, seek medical help immediately.\n3. Immobilize the leg: Use a splint, sling, or other available materials to immobilize the leg and prevent movement. Be sure not to apply too much pressure on the injury site.\n4. Provide pain relief: Offer over-the-counter pain medication, such as acetaminophen or ibuprofen, to help manage pain.\n5. Seek medical attention: If the fracture is severe or if you suspect that there may be other injuries, seek medical help as soon as possible.\n\nOnce you've ensured the person's safety and stabil
```

Strengths	Opportunities
• Strong emphasis on first aid – critical early actions • Uses an ordered list that is easy to follow • The answers provides simple non-technical wording usable for general audiences • The response is practical and actionable	• The 125-token cap restricts the ability to fully elaborate on complex protocols, leading to oversimplification. • Lacks detail on proper immobilization techniques

Data Preparation for RAG

1 **Loading the Data**	`manual_pdf_path = "medical_diagnosis_manual.pdf"`
	`pdf_loader = PyMuPDFLoader(manual_pdf_path)`
	`manual = pdf_loader.load()`
		This code initializes a document ingestion step for the file "**medical_diagnosis_manual.pdf**." It defines the file path, uses PyMuPDFLoader to parse the document, and executes .load() to extract text into structured page-level objects with metadata.

2 **Data Chunking**	`from langchain.text_splitter import RecursiveCharacterTextSplitter`
	`text_splitter = RecursiveCharacterTextSplitter.from_tiktoken_encoder(`
	`    encoding_name='cl100k_base',`
	`    chunk_size=500, #Complete the code to define the chunk size`
	`    chunk_overlap= 25  #Complete the code to define the chunk overlap`
	`)`
	`document_chunks = pdf_loader.load_and_split(text_splitter)`
	`len(document_chunks)`
	`8680`
		This code processes "medical_diagnosis_manual.pdf" into structured, model-ready chunks using RecursiveCharacterTextSplitter with the *cl100k_base* tokenizer.
		
		A chunk size of 500 and overlap of 25 balance context retention and efficiency, improving embedding quality and retrieval accuracy. The load_and_split method combines ingestion and segmentation.
		
		The result (8,680 chunks) enables scalable semantic search and improved performance in RAG applications.

Data Preparation for RAG

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Embedding

```python
from langchain_community.embeddings import SentenceTransformerEmbeddings
```

```python
embedding_model = SentenceTransformerEmbeddings(model_name="all-MiniLM-L6-v2")
```

This code initializes an embedding layer using SentenceTransformerEmbeddings with the **"all-MiniLM-L6-v2"** model to convert text into numerical vector representations. These embeddings capture semantic meaning, enabling similarity comparisons across documents.

The selected model is lightweight and efficient, producing 384-dimensional vectors

4

Vector Database

```python
vectorstore = Chroma.from_documents(
    document_chunks, #Complete the code to pass the document chunks
    embedding_model, #Complete the code to pass the embedding model
     persist_directory=out_dir
)
```

```python
vectorstore = Chroma(persist_directory=out_dir,embedding_function=embedding_model)
```

This code initializes a persistent vector database using Chroma to store and manage embeddings generated from document chunks. The from_documents method ingests processed text chunks alongside the embedding model, converting them into indexed vectors for efficient retrieval.

Data Preparation for RAG

5

Retriever	`retriever = vectorstore.as_retriever(`
	`search_type='similarity',`
	`search_kwargs={'k':3}`
		This code configures a retriever from the Chroma vector store to enable efficient semantic search over embedded document chunks. Using search_type='similarity', it retrieves results based on vector similarity to a given query, ensuring relevant contextual matches.
		
		The parameter k=3 limits output to the top three most similar documents, balancing relevance with response efficiency. This focused retrieval supports high-quality context injection into LLM prompts.

Question Answering for RAG

RAG Model Setup

Data Preparation for RAG

1. Loading the Data
2. Data Chunking
3. Embedding
4. Vector Database
5. Retriever

System Message

qna_system_message = """You are an AI Medical Assistant that provides accurate and helpful medical information.

Please answer only using the provided context. Do not add or infer anything beyond the context.

If the answer is not in the context, say you do not know.

User Prompt Template

###Context

Here are some documents that are relevant to the question mentioned below.

{context} ###Question {question}

Response Function

- **k**= 3
- **max_tokens** = 128
- **temperature** = 0 (0: fully deterministic - 1: creative)
- **top_p** = 0.95 (nucleus sampling)
- **top_k** = 50 (50 most likely words)

- The process begins with data preparation, including loading, chunking, and embedding documents, which are then stored in a vector database for efficient semantic retrieval. A retriever identifies the top relevant documents (k=3) to support each query.
- The system is governed by a strict system message, positioning the model as a medical assistant that must rely on the **medical_diagnosis_manual.pdf** . The system message explicitly restricts hallucinations by instructing the model to avoid inference beyond retrieved data and to acknowledge when information is unavailable.
- The response function parameters are optimized for reliability and evaluation: low temperature (0) ensures deterministic outputs, while top_p (0.95) and top_k (50) balance coherence and token selection. A max token limit of 128 controls response length.

Question Answering for RAG

Query 1: What is the protocol for managing sepsis in a critical care unit?

Llama.generate: prefix-match hit
Based on the context provided, the protocol for managing sepsis in a critical care unit includes:\n1. Suspecting sepsis or septic shock based on symptoms such as shaking chills, persistent fever, altered sensorium, hypotension, and GI symptoms (abdominal pain, nausea, vomiting, diarrhea).\n2. Obtaining cultures of blood and any other appropriate specimens for diagnosis.\n3. Administering empiric antibiotics after obtaining appropriate cultures.\n4. Continuing therapy by adjusting antibiotics according to culture and suscept

Strengths	Opportunities
• The response incorporates mote precise content showing an improvement in clinical accuracy • Better articulation of treatment steps • Stronger alignment to clinical protocols • RAG significantly enhances accuracy, depth, and reliability, transforming the response from general guidance to more clinically grounded insight	• Clinical specificity can be expanded • The response is limited by token constrains • Enhancing protocol completeness and strengthening time-based prioritization would improve the clinical rigor and overall medical reliability of the response.

Question Answering for RAG

Query 2: What are the common symptoms for appendicitis, and can it be cured via medicine? If not, what surgical procedure should be followed to treat it?
Llama.generate: prefix-match hit Based on the context provided, the common symptoms for appendicitis include epigastric or periumbilical pain followed by brief nausea, vomiting, a norexia, and a shift in pain to the right lower quadrant. Pain increases with cough and motion, and there may be direct and rebound tenderness at M cBurney's point. Other possible signs include Rovsing sign, psoas sign, obturator sign, low-grade fever, and less frequent or absent bowel moveme nts. However, these classic findings appear in less than 50% of patients, and

Strengths	Opportunities
• Improved clinical accuracy • More precise identification of appendicitis symptoms and typical progression of pain • The answer covers symptoms, diagnosis and treatment in a logical and organized manner • Enhanced medical releveance	• The answer could expand on surgical approaches and recovery expectations • Does not differentiate when antibiotics alone are appropriate vs when surgery is required • Procedural detail and urgency emphasis would strengthen the effectiveness of the answer.

Question Answering for RAG

Query 3: What are the effective treatments or solutions for addressing sudden patchy hair loss, commonly seen as localized bald spots on the scalp, and what could be the possible causes behind it?

Llama.generate: prefix-match hit
Based on the context provided, the condition being described is Alopecia Areata. The treatment options mentioned in the text include:\n1. Oral anti malarials\n2. Corticosteroids\n3. Retinoids\n4. Immunosuppressants\nThese treatments are indicated for Lichen Planopilaris and chronic cutaneous lupus lesions, but they can also be used for Alopecia Areata in some cases. The text suggests that Alopecia Areata is an autoimmune disorder affec ting genetically susceptible people exposed to unclear environmental triggers

Strengths	Opportunities
• Improved clinical grounding – more accurate identification of causes • The answer includes more actionable treatment aligned with dermatological practices • The answer maintains clear organization across causes, symptoms and treatments	• Limited treatment specificity • The response is limited by token constrains • the response would benefit from greater clinical specificity, diagnostic detail, and prioritization of causes. • Minimal coverage of diagnosis protocols

Question Answering for RAG

Query 4: What treatments are recommended for a person who has sustained a physical injury to brain tissue, resulting in temporary or permanent impairment of brain function?
Llama.generate: prefix-match hit ###Answer\nThe documents recommend supportive care as the primary treatment for a person with brain damage. This includes preventing system ic complications such as pneumonia, UTI, and thromboembolic disease, providing good nutrition, and preventing pressure ulcers. Early intervention by rehabilitation specialists is also indispensable for maximal functional recovery. Rehabilitation therapy varies depending on the patient's abnormali ties and includes prevention of secondary disabilities, such as pressure ulcers and joint contractures, and family education. For patients whose com a exceeds 24 hours, a prolonged period of re

Strengths	Opportunities
• The answer provides comprehensive care coverage. • Very strong clinical accuracy –the answer provides aligned treatment approaches for brain injury management. • The answer is well grounded in retrieved medical knowledge.	• Surgical interventions are mentioned but not explained. • The response would benefit from severity-based differentiations. • The answer doesn't differentiate between immediate vs long term interventions.

Question Answering for RAG

Query 5: What are the necessary precautions and treatment steps for a person who has fractured their leg during a hiking trip, and what should be considered for their care and recovery?

Llama.generate: prefix-match hit
Based on the context provided, here's the answer:\n1. Evaluation in the emergency department: The person should be evaluated for signs or sympt oms of hemorrhagic shock due to occult blood loss, as well as for ischemia (absent pulses, marked pallor, coolness distal to the injury, severe pain). \n2. Immobilization: The leg should be immobilized immediately by splinting to prevent further injury and decrease pain. This can be done with a non rigid or noncircumferential device.\n3. Definitive treatment: Dep

Strengths	Opportunities
• The answer incorporate reliable first-aid and fracture management principles. • Clear step-based structure provides guidance supports in high stress environments. • The answer addresses immediate precautions and touches on recovery considerations.	• The response would benefit from greater clinical detail, emergency prioritization, and deeper recovery guidance • The response is limited by token constrains • Limited detail on rehabilitation, healing timelines, and weight-bearing progression

Question Answering for RAG w/. Tunning

RAG Model Setup

Data Preparation for RAG

1. Loading the Data
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User Prompt Template

###Context

Here are some documents that are relevant to the question mentioned below.

{context} ###Question {question}

Response Function w/. Tunning

- **k**= 3
- **max_tokens** = 128
- **temperature** = 0.5 <Tuned>
- **top_p** = 0.95
- **top_k** = 50

- This Retrieval-Augmented Generation (RAG) framework enhances a locally deployed LLM by grounding responses in retrieved medical knowledge.
- The pipeline includes data preparation (loading, chunking, embeddings) and storage in a vector database, with a retriever (k=3) selecting the most relevant context.
- A controlled system prompt defines the model as an AI Medical Assistant and enforces strict guardrails—requiring use of only provided context and acknowledging unknowns, reducing hallucination risk.
- The user prompt template injects context into each query, enabling grounded responses. Parameter tuning (temperature = 0.5, top-p = 0.95, top-k = 50, max tokens = 128) balances consistency and flexibility while maintaining concise outputs.
- Overall, this setup improves accuracy, control, and reliability, enabling consistent, context-driven medical responses.

Question Answering for RAG w/. Tunning

Query 1: What is the protocol for managing sepsis in a critical care unit?
Llama.generate: prefix-match hit Based on the provided context, the protocol for managing sepsis in a critical care unit includes:\n1. Suspected cases should undergo cultures of blo od and any other appropriate specimens.\n2. Empiric antibiotics are given to patients with suspected bacteremia while awaiting culture results.\n3. Antibiotic therapy is adjusted based on the results of culture and susceptibility testing, surgical drainage of abscesses, and removal of internal devic es that may be the source of bacteria.\n4. Close monitoring of vital signs and other physiologic parameters with ongoing and continuous

Strengths	Opportunities
• High Clinical accuracy – The answer incorporates guideline aligned sepsis management. • Temperature of 0.5 allows for a more natural response while maintaining clinical grounding. • The answer provides more actionable steps aligned with ICU protocols.	• The answer could improve with a step by stem time sequenced and prioritization. • The response is limited by token constrains

Question Answering for RAG w/. Tunning

Query 2: What are the common symptoms for appendicitis, and can it be cured via medicine? If not, what surgical procedure should be followed to treat it?
Llama.generate: prefix-match hit The common symptoms for appendicitis include epigastric or periumbilical pain followed by brief nausea, vomiting, and anorexia; after a few hours, t he pain shifts to the right lower quadrant. Pain increases with cough and motion. Classic signs are direct and rebound tenderness located at McBur ney's point (junction of the middle and outer thirds of the line joining the umbilicus to the anterior superior spine). Additional signs include pain felt in the right lower quadrant with palpation of the left lower quadrant (Rovsing sign), an

Strengths	Opportunities
• The answer is grounded in retrieved medical content, increasing reliability. • The answer has a logical flow across symptoms, diagnosis and treatment options. • Temperature 0.5 improves readability while maintaining consistency	• Doesn’t clearly define when antibiotics alone are appropriate vs surgery • Minimal detail on imagining and lab tests • The response is limited by token constrains

Question Answering for RAG w/. Tunning

Query 3: What are the effective treatments or solutions for addressing sudden patchy hair loss, commonly seen as localized bald spots on the scalp, and what could be the possible causes behind it?

Llama.generate: prefix-match hit
Based on the context provided, the condition being described is alopecia areata, a type of nonscarring alopecia characterized by sudden, patchy hai r loss. The most common treatments for addressing this condition include:\n\n1. Topical minoxidil: This medication prolongs the anagen growth pha se and enlarges miniaturized follicles in male-pattern or female-pattern hair loss. However, it is not effective for all causes of hair loss, including alop ecia areata.\n2. Oral antimalarials, corticosteroids, retinoids

Strengths	Opportunities
• The answer directly answer the question without unnecessary information • The answer is grounded in retrieved context, improving factual reliability. • Correctly identifies alopecia areata as the likely condition based on symptoms	• Does not fully cover standard therapies • Lack of structure – doesn't follow the predefined sections (causes, symptoms, diagnosis. Treatment)

Question Answering for RAG w/. Tunning

Query 4: What treatments are recommended for a person who has sustained a physical injury to brain tissue, resulting in temporary or permanent impairment of brain function?
Llama.generate: prefix-match hit Based on the context provided, the following treatments are recommended for a person who has sustained a physical injury to brain tissue:\n1. Supportive care: This includes preventing systemic complications due to immobilization (such as pneumonia, UTI, and thromboembolic disease), providing good nutrition, and preventing pressure ulcers.\n2. Rehabilitation therapy: Early intervention by rehabilitation specialists is essential for maximal functional recovery. Intervention includes prevention of secondary disabilities (such as pressure ulcers, joint contractures), prevention of pneumonia, and family education. Patients should be

Strengths	Opportunities
• The response includes emergency care, medications and rehabilitation. • The organized format enhances comprehension and usability. • The answer reflects more reliable evidence approach to brain injury management.	• Surgical interventions are mentioned but not explained in detail. • Missing ICU specific and details. • The response is limited by token constrains. • The answer lacks prioritization steps and patient guidance.

Question Answering for RAG w/. Tunning

Query 5: What are the necessary precautions and treatment steps for a person who has fractured their leg during a hiking trip, and what should be considered for their care and recovery?
Llama.generate: prefix-match hit Based on the context provided, here's the answer:\n\n1. Evaluation in the emergency department: The person should be evaluated for serious injuri es to all organ systems and resuscitated if necessary. They should also be checked for symptoms or signs of ischemia (absent pulses, marked pall or, coolness distal to the injury, severe pain) due to potential blood loss from the fracture.\n2. Immobilization: The leg should be immobilized immedi ately by splinting to prevent further injury and decrease pain. This can be done with a nonrigid or noncircum

Strengths	Opportunities
• The answer provides reliable first-aid guidance • The answer is grounded in retrieved context, improving factual reliability. • The answer provides a clear step-based structure. • Tem 0.5 supports clarity while maintaining consistency.	• The answer doesn’t distinguish fracture types. • The answer lacks detail around pain management and splitting techniques. • The response is limited by token constrains.

Output Evaluation

RAG Self Evaluation Parameters

Parameter 1: Groundedness System Message

"""You are an AI assistant that evaluates whether an answer is grounded in the provided context.

If the answer is fully supported by the context, respond with 'Grounded'.

If the answer contains information not supported by the context, respond with 'Not Grounded'.

Only respond with 'Grounded' or 'Not Grounded'."""

Parameter 2: Relevance System Message

"""You are an AI assistant that evaluates whether the provided context is relevant to the given question.

If the context is relevant to answering the question, respond with 'Relevant'.

If the context is not relevant to the question, respond with 'Not Relevant'.

Only respond with 'Relevant' or 'Not Relevant'."""

Response Function

- **k**= 3
- **max_tokens** = 128
- **temperature** = 0
- **top_p** = 0.95
- **top_k** = 50

- This framework defines a structured evaluation approach for a Retrieval-Augmented Generation (RAG) system, centered on **groundedness** and **relevance**.
- This model configuration leverages a locally deployed **Mistral 7B Instruct** model optimized for deterministic and consistent output generation.
- **Groundedness** ensures responses are fully supported by retrieved context, labeled "Grounded" or "Not Grounded" to reduce hallucinations.
- **Relevance** evaluates whether retrieved content addresses the query, labeled "Relevant" or "Not Relevant".
- Configuration uses k=3, max_tokens=128, temperature=0, top_p=0.95, and top_k=50 for consistent, controlled outputs.
- Overall, this approach improves RAG reliability by enforcing factual accuracy and contextual alignment, strengthening trust and decision-making.

Output Evaluation

Query 1: What is the protocol for managing sepsis in a critical care unit?
Grounded. The context provides information on the signs of sepsis or septic shock and the steps to take in managing such conditions, including obtaining cultures and administering empiric antibiotics. **Relevant.** The context discusses the diagnosis and treatment of sepsis in a critically ill patient, which is directly related to the question about managing sepsis in a critical care unit.

Evaluation
• The evaluation indicates the response is both **grounded** and **relevant**, demonstrating strong alignment between the generated answer and the retrieved medical source content. The supporting context appropriately references core sepsis management protocols in a critical care setting, including recognition of sepsis or septic shock, obtaining cultures, and initiating empiric antibiotic therapy.

Output Evaluation

Query 2: What are the common symptoms for appendicitis, and can it be cured via medicine? If not, what surgical procedure should be followed to treat it?
Grounded. The context provides detailed information about the symptoms of appendicitis, the fact that it cannot be cured with medicine alone, and the standard treatment of surgical removal. **Relevant.** The context provides detailed information about the symptoms, diagnosis, and treatment for appendicitis, which directly answers the question.

Evaluation

- The evaluation confirms the response is both **grounded** and **relevant**, reflecting strong alignment between the generated output and the retrieved medical source material. The supporting context appropriately outlines common appendicitis symptoms, clarifies that medication alone is typically not sufficient, and identifies surgical removal as the primary treatment standard.

Output Evaluation

Query 3: What are the effective treatments or solutions for addressing sudden patchy hair loss, commonly seen as localized bald spots on the scalp, and what could be the possible causes behind it?

Grounded. The context provides information on alopecia areata, its causes, and treatments.

Relevant. The context provides information on various causes of hair loss, including alopecia areata, and discusses possible treatments for this condition.

Evaluation

- The evaluation confirms the response is both **grounded** and **relevant**, demonstrating strong alignment between the generated output and the retrieved medical source content. The supporting context appropriately identifies alopecia areata as a likely cause of sudden patchy hair loss and includes relevant information on underlying causes and available treatment options.

Output Evaluation

Query 4: What treatments are recommended for a person who has sustained a physical injury to brain tissue, resulting in temporary or permanent impairment of brain function?

Grounded. The context mentions supportive care, prevention of systemic complications, good nutrition, preventing pressure ulcers, and early intervention by rehabilitation specialists for brain damage patients. The answer expands on these points and adds information about the types of rehabilitation therapy that may be required.

Relevant. The context discusses various treatments for brain damage, including supportive care and rehabilitation therapy.

Evaluation

- The evaluation confirms the response is both **grounded** and **relevant**, demonstrating strong alignment between the generated output and the retrieved medical source content. The supporting context appropriately addresses treatment approaches for brain injury, including supportive care, prevention of secondary complications, nutritional management, pressure ulcer prevention, and early involvement of rehabilitation specialists.

Output Evaluation

Query 5: What are the necessary precautions and treatment steps for a person who has fractured their leg during a hiking trip, and what should be considered for their care and recovery?

Grounded. The context provides detailed information on the necessary precautions and treatment steps for a person who has fractured their leg, including immobilization, pain management, definitive treatment, RICE therapy, and rehabilitation.

Relevant. The context provides information on the necessary precautions and treatment steps for a person who has fractured their leg, including immobilization, pain management, definitive treatment, RICE therapy, and rehabilitation.

Evaluation

- The evaluation confirms the response is both **grounded** and **relevant**, demonstrating strong alignment between the generated output and the retrieved medical source content. The supporting context appropriately addresses key fracture management steps, including immobilization, pain control, definitive treatment, RICE therapy, and rehabilitation considerations for recovery.

APPENDIX

Data Background and Content

File Content

File Information

- **File**: medical_diagnosis_manual.pdf
- **Title:** The Merck Manual of Diagnosis and Therapy – 19th Edition (2011)
- **File Description:** The Merck Manual of Diagnosis and Therapy is one of the most widely respected and comprehensive medical reference resources used by healthcare professional worldwide
- **Size**: 19.2 MB

File Layout

Page Number : 3
Table of Contents

Number of Pages

- **len(manual):** 4114

Happy Learning !